Seafood Alert: Final Project Design Doc

Romaan Shaikh, CS 19300

1.1 Problem Statement

Seafood is one of the few proteins I can reliably eat in Purdue dining courts, but I never know when or where it is being served without manually checking every menu every day. Even when shrimp or fish shows up, some dishes are prepared with meat products like chicken base or chicken fat, so I have to dig through ingredient lists to know if it is actually safe for me. I care about this because it affects what I eat every single day.

1.2 Idea (The Solution)

A Java program that checks every Purdue dining court menu each day using the Purdue Dining API, finds seafood dishes that contain no meat ingredients, and sends me a Telegram message on my phone listing what is being served, where, and at which meal.

2.1 Eisenhower Matrix

	Urgent	Not Urgent
Important	**Do (MVP)** ● Pull each dining court's daily menu from the Purdue Dining API ● Detect seafood items using the Fish and Shellfish allergen tags ● Filter out items whose ingredients contain meat (chicken base, chicken fat, beef, pork, gelatin) ● Send a Telegram notification with item, dining court, and meal ● Track already notified items in a text file to avoid duplicate pings	**Schedule (V2.0+)** ● Weekly lookahead that pings for the whole upcoming week ● Integrate into my existing personal food app ● Recommend the best time and location when multiple protein options exist ● User config file for custom keywords and favorite dining courts
Not Important	**Delegate:** ● JSON parsing (Gson) ● HTTP requests (java.net.http.HttpClient) ● Daily scheduling (cron) ● Message delivery	**Delete** ● Full weekly calorie and macro tracking ● Comparing dining court, meal swipe, and dining dollar options for value

	Urgent	Not Urgent
	(Telegram Bot API)	and variety ● Modeling every restaurant's weekend hours and closures ● Graphical user interface

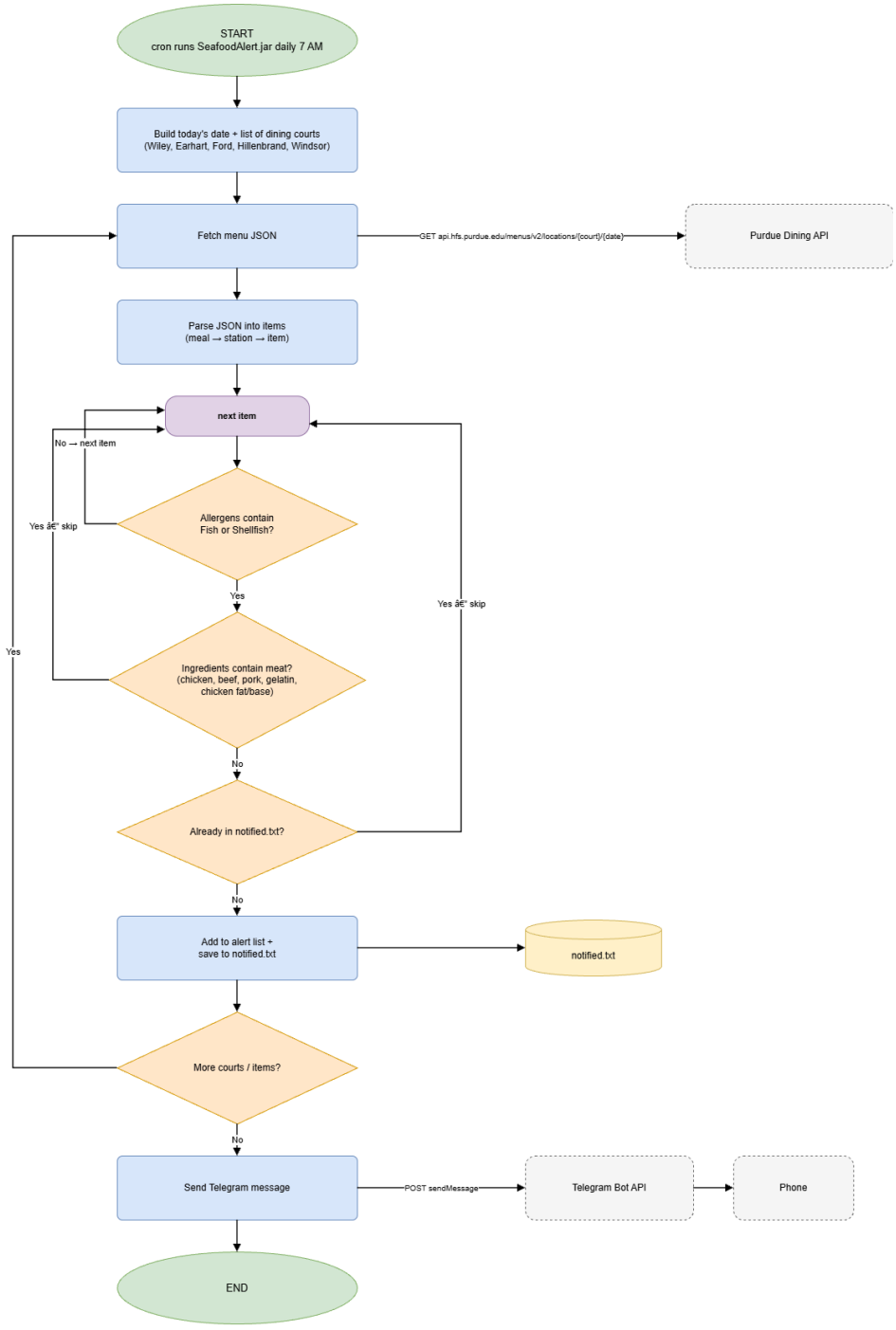
2.2 Tech Stack

- Language: Java 17+
- HTTP: java.net.http.HttpClient (built into Java)
- JSON parsing: Gson library
- Data source: Purdue Dining API, https://api.hfs.purdue.edu/menus/v2/locations/{location}/{date} (free, public, no API key, returns JSON)
- Item details: https://api.hfs.purdue.edu/menus/v2/items/{itemId} for ingredient lists
- Notifications: Telegram Bot API (sendMessage endpoint, bot token stored in a local config file that is not committed)
- Data storage: notified.txt plain text file of item IDs and dates already sent
- Scheduling: cron job runs the program once every morning
- Version control: Git and GitHub

2.3 Chunked Build Plan

1. Set up GitHub repo, folder structure, and README
2. Fetch one dining court's menu from the API and print the raw JSON
3. Parse the JSON with Gson and print every item name by meal and station
4. Filter items to only those with Fish or Shellfish allergens
5. Fetch ingredients for each seafood item and remove anything containing meat
6. Loop through all dining courts and collect the results
7. Create a Telegram bot and send a test message from Java
8. Send the filtered seafood list as a formatted Telegram message
9. Save notified items to notified.txt and skip duplicates
10. Schedule the program with cron and test a full daily run

2.4 System Diagram



3.1 GitHub Repository

https://github.com/istaawoo/seafood-alert